

RAJITHA B

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EDUCATION

Chennai Institute Of Technology

B.E. Computer Science and Engineering

Expected May 2027

PROJECTS

Real-Time Text2Sign (ISL) & Sign-to-Speech System

Python | OpenCV | MediaPipe

- Built a real-time gesture recognition pipeline using OpenCV and MediaPipe, **extracting 21-point landmarks at 25 to 30 FPS**; designed a custom CNN-LSTM architecture for temporal sequence modeling.
- Developed a data pipeline for collection, preprocessing, and labeling of 1,000+ samples, **achieving 90% accuracy with F1-score evaluation**; implemented text ISL speech bidirectional mapping.
- Optimized inference latency to <100ms via lightweight model tuning and **edge-device deployment** (Raspberry Pi); enabled real-time prediction with efficient memory and compute usage.

Distributed Code Execution & Evaluation Platform

React.js | Node.js | Docker | Redis | REST APIs

- Built a multi-language code execution service using **Docker-based sandboxing**, isolating runtime environments and enforcing resource limits for secure execution.
- Implemented a Redis-backed job queue with worker processes, **handling 500+ concurrent executions** with retry logic and efficient scheduling.
- Developed test case orchestration and performance tracking, capturing execution time/memory metrics and exposing results via REST APIs for real-time feedback.

INTERNSHIP

CubeSat intern - ULOG3, Chennai.

Nov 2023 - Apr 2024

Python | Deep Learning | OpenCV | Data Preprocessing | Model Evaluation

- Built image processing pipelines for enhancement and object detection using OpenCV, **achieving 85% accuracy on experimental datasets**.
- Trained and evaluated deep learning models for vision tasks and **integrated Raspberry Pi camera modules** for real-time data acquisition in embedded setups.
- Structured data preprocessing and evaluation workflows, improving model consistency and reducing experimentation time during iterative testing.

SSV Traders & InternPe Web Developer Intern

May 2024 - June 2024

React.js | HTML5 | CSS3 | Frontend Development

- Developed responsive React.js interfaces with component-based architecture, **improving page load performance by 25%** and ensuring cross-device compatibility.
- Recreated the Ninjacart platform through frontend cloning, focusing on layout structuring, routing, and reusable UI components.
- Enhanced UI interactivity and usability by implementing dynamic elements and optimized rendering, improving overall user engagement.

SKILLS

Languages: Python, Java, JavaScript, SQL

Frontend: React.js, HTML, CSS

Backend: Node.js, Express.js, REST APIs

AI/ML & Computer Vision: OpenCV, MediaPipe, PyTorch, TensorFlow

Databases & Tools: MongoDB, Redis, Docker, Git, VS Code

ACHIEVEMENTS & LEADERSHIP

- Solved 500+ DSA problems** across coding platforms with strong problem-solving fundamentals.
- Achieved **global rank 1849 in LeetCode** contest competitions.
- Collaborated effectively in team-based project development and technical problem-solving environments.

COURSE

CCNA (Networks) | Programming Essentials in Python

MongoDB Basics | Generative AI Fundamentals